

Characterizing the Performance of Interstate Flexible Pavements Using Artificial Neural Networks and Random Parameters Regression

Mohamed S. Yamany, S.M.ASCE¹; Tariq Usman Saeed, A.M.ASCE²; Matthew Volovski³; and Anwaar Ahmed, A.M.ASCE⁴

Abstract: Past studies developed pavement performance models using data from all or multiple states across the United States. This study hypothesized that due to variation in agency practices and work activity profiles, individual pavement performance models should be estimated for each state, using data from its own roadway network, for use in its pavement management system. To test this hypothesis, this study used condition data of Interstate flexible pavements from eight Midwestern states to estimate three models: fixed-parameters regression, random-parameters regression, and artificial neural networks (ANNs). The ANNs model was found to statistically outperform the regression counterparts when estimating pavement roughness across all states. In contrast, the random-parameters model was statistically superior to the ANNs model in some cases when exploring the performance of these models for individual states. The statistical performance of models did not have a consistent trend across all states. Therefore, the application of models, based on data from multiple jurisdictions, could lead to erroneous/nonoptimal maintenance and rehabilitation decisions. Highway agencies are recommended to rely on their own jurisdictional data when developing their pavement performance models. DOI: 10.1061/(ASCE)IS.1943-555X.0000542. © 2020 American Society of Civil Engineers.

Author keywords: Interstate flexible pavement; Performance models; Artificial neural networks; Random-parameters regression.

Introduction

In the United States, more than four million miles of roadways provide mobility, accessibility, and connectivity throughout the country, and play a significant role in improving the overall quality of life, fostering economic growth, and enhancing national security. Therefore, maintaining pavements in a state of good repair is of paramount importance for state highway agencies (SHAs) and other related agencies. Pavement deterioration is characterized by surface distresses and structural deficiencies, which mainly occur due to repeated traffic and climatic loadings. Assessing the present condition and predicting the future performance of pavements are critical elements of a pavement management system. In this regard, pavement performance models are imperative for several agency activities including maintenance programming and scheduling.

For a comprehensive review of past research related to the objectives of this paper, Table 1 summarizes past studies of pavement

performance with respect to study region, pavement surface type, pavement condition indicator, highway functional class, data source, modeling methodology, and explanatory variables. Most past studies developed pavement performance models using the Long-Term Pavement Performance (LTPP) data or local agency data. The most frequently used modeling techniques include regression and machine learning, especially artificial neural networks (ANNs).

The performance of pavements is identified by measuring and observing their condition over their lifetime. Several types of indexes or indicators are used to characterize pavement condition. Pavement condition indexes can represent pavement structural condition (e.g., pavement structural number or distress score), pavement functional condition [e.g., International Roughness Index (IRI) or riding quality], or both pavement structural and functional conditions [e.g., pavement condition index (PCI) or pavement condition rating (PCR)]. Each SHA in the United States uses different pavement condition indexes based on its policy and pavement management system. For instance, the state of Ohio uses the PCR (Rajagopal 2006), whereas, the states of Indiana, Illinois, Connecticut, Colorado, and California use the IRI (Bektas et al. 2014). The IRI, PCR, and rutting (RUT) were used to measure the effectiveness of rehabilitation treatments for the state of Indiana in a study by Labi et al. (2006), and to predict pavement condition in studies by Sarwar and Anastasopoulos (2016, 2017) and David et al. (2017). The majority of SHAs and past studies (Table 1) used the IRI to measure the surface condition of hot-mix asphalt (HMA) or flexible pavements.

Past studies found several factors that affect the deterioration of pavement performance; these include pavement age, climatic conditions, and traffic loadings (Table 1). Pavement age has been recognized as the most statistically significant variable in predicting pavement performance (Abaza 2004; Kim and Kim 2006; Rajagopal 2006). Ahmed et al. (2016) used traffic loading and climate conditions to predict pavement deterioration. Other variables that were employed to predict pavement performance include

¹Doctoral Candidate, Lyles School of Civil Engineering, Purdue Univ., West Lafayette, IN 47907 (corresponding author). ORCID: <https://orcid.org/0000-0002-7828-6075>. Email: myamany@purdue.edu

²Research Affiliate, Center for Connected and Automated Transportation, Lyles School of Civil Engineering, Purdue Univ., West Lafayette, IN 47907. ORCID: <https://orcid.org/0000-0003-0752-9217>. Email: tusaeed@purdue.edu

³Assistant Professor, Dept. of Civil and Environmental Engineering, Manhattan College, 4513 Manhattan College Pkwy., Riverdale, NY 10471. ORCID: <https://orcid.org/0000-0002-8222-4600>. Email: mvolovski01@manhattan.edu

⁴Associate Professor, National Univ. of Science and Technology, Risalpur, NUST Main Campus, H-12, Islamabad, Pakistan. ORCID: <https://orcid.org/0000-0002-5280-3652>. Email: anwaar5148@gmail.com

Note. This manuscript was submitted on June 5, 2019; approved on November 26, 2019; published online on March 18, 2020. Discussion period open until August 18, 2020; separate discussions must be submitted for individual papers. This paper is part of the *Journal of Infrastructure Systems*, © ASCE, ISSN 1076-0342.

Table 1. Summary of insights from past research

Author(s) (year)	Study region	Country	Pavement surface type	Condition indicator	Road class	Data source	Modeling methodology	Explanatory variables
Abaza (2004)	—	United States	AC	PSI	—	AASHTO	FP	PA, MR, ESALs, SN
Abdelaziz et al. (2020)	—	United States, Canada	AC	IRI	—	LTPP	FP, ANN	PA, initial IRI, transverse and alligator cracks, RUT
Ahmed et al. (2016)	Indiana	United States	AC, PCC	IRI	IS, NHS-NIS	INDIPAVE2000, InDoT (2011)	FP	AADT, AAFI, time
Aleadelat et al. (2018)	Wyoming	United States	AC	PSI	Locals	Pathway services 2014	FP	PCI, IRI, RUT
Al-Suleiman and Shiyab (2003)	Dubai	UAE	AC	IRI	Majors	In situ gathered data	FP	PA, PSI
Amin and Amador-Jiménez (2016)	Montreal city	Canada	AC, PCC	PCI	Collectors, arterials	Montreal city database	ANN (BPN, GDR algorithms)	AADT, ESALs, SN, PA, layer thickness
Bektas et al. (2014)	Iowa	United States	AC, PCC, others	PCI-2	IS, Non-NHS	Iowa database	FP	Cracking, riding, faulting indexes
Bianchini and Bandini (2010)	Minnesota	United States (Midwest)	AC	PSI	Rural	MnRoad	ANN/feedforward, Fuzzy Logic	SCI, deflection RUT, layer thickness, ESALs
Dalla Rosa et al. (2017)	Texas	United States	AC	IRI	Rural, Urban	TxDOT database	FP	PA, initial IRI, climate zone, ESALs, layer thickness
Hong and Prozzi (2010)	Minnesota	United States (Midwest)	AC	IRI	Rural	MnRoad	FP, RP, RE	ESALs, layer thickness, PA, frost heave, maintenance, asphalt mixture
Khan et al. (2014)	Queensland	Australia	AC	DR	—	Queensland (TMR-QLD database)	Markov chain	PA, layer thickness, AADT, ESALs
Khattak et al. (2014)	Louisiana	United States	Composite	IRI	IS, arterials, collectors	Louisiana DOT database	FP	ESALs, pre- and post-treatment IRI, PI, TI
Kim and Kim (2006)	Georgia	United States	AC	PACES	State, IS	PACES database of Georgia	FP	PA, AADT
La Torre et al. (1998)	—	United States	AC	IRI	—	LTPP	ANN	ESALs, freezing index, air voids, base thickness, precipitation, PA
Labi et al. (2006)	Indiana	United States (Midwest)	AC	IRI, PCR, RUT	IS, NIS	INDOT, Indiana County Flow Map, NOAA	FP	AADT, %truck, ESALs, freeze thaw cycles, freezing index, micro-surfacing treatment
Lou et al. (2001)	Florida	United States	AC, PCC	CI	—	FDOT database	ANN (BPN)	Maintenance cycle, PA
Luo (2013)	Ohio	United States (Midwest)	AC	PCR	—	Ohio DOT database	Autoregression	PA, past pavement condition
Mazari and Rodriguez (2016)	Multiple regions ^a	United States, Canada	AC	IRI	—	LTPP	GEP, ANN, Hybrid GEP and ANN	PA, ESALs, initial IRI
Eldin and Senouci (2011)	Oregon	United States	AC	CR	—	Oregon DOT database	ANN (BPN)	RUT, bleeding, alligator and transverse, and block cracking, patching/pothole, raveling
Plati et al. (2016)	—	Greece	AC	ϵ	—	In situ gathered data	FP, ANN	Deflection, layer thickness
Prozzi and Madanat (2003)	—	United States	AC	PSI	—	AASHTO Road Test 1962	FP, RE	ESALs, frost gradient, layer thickness
Rajagopal (2006)	Ohio	United States (Midwest)	AC, PCC	PCN	Urban majors, minors	Cincinnati database	FP	PA, surface condition, environmental factors
Rose et al. (2016)	—	India	AC	Raveling, pothole, edge failure	Rural	Binu (2012)	Probabilistic	PA, construction quality
Saeed et al. (2017)	—	—	Multiple types	—	Multiple classes	—	FP	Traffic loading, freezing index

Table 1. (Continued.)

Author(s) (year)	Study region	Country	Pavement surface type	Condition indicator	Road class	Data source	Modeling methodology	Explanatory variables
Sandra and Sarkar (2013)	Rajasthan state	India	AC	IRI	NHS, state, major district	In situ gathered data	FP	Cracking, potholes, patching, RUT, raveling
Silva et al. (2000)	Michigan	United States (Midwest)	AC	PASER	—	Michigan counties	Logistic growth, Markov chain	PA
Terzi (2007)	—	United States	AC	PSI	—	AASHTO	ANN (LMBPN)	IRI, cracking, patching, RUT
Xu et al. (2015)	Kentucky	United States	AC	IRI, cracking, raveling	IS parkways	KTC	FP	AADT, PA, cracking, raveling, IRI
Yang et al. (2003)	Florida	United States	AC, PCC	Cracking, riding, RUT	—	FDOT database	ANN (BPN)	PA, maintenance cycle
Ziari et al. (2016)	—	United States	AC	IRI	—	LTPP	ANN, GMDH	PA, AAP, AADT, AAFI, AAT, AADTT, ESALS, ST, PT

Note: CI = crack index; BPN = backpropagation neural; FDOT = Florida department of transportation; CR = condition rating; PSI = present serviceability index; LMBPN = Levenberg–Marquardt back-propagation neural; GMDH = group method of data handling; AAP = annual average precipitation; AAT = annual average temperature; AAFI = annual average freezing index; ESALS = equivalent single axle loads; AADT = annual average daily traffic; AADTT = annual average daily truck traffic; TxDOT = Texas Department of Transportation; GEP = gene expression programming; SN = structural number; PASER = pavement surface evaluation and rating; MnRoad = database of Minnesota road test project; RE = random-effects model; RP = random-parameter regression; NP = fixed-parameter regression; NHS = national highway system; Non-NHS = non-national highway system; PCI-2 = pavement condition index developed for Iowa state; FWD = falling weight deflectometer; SCI = surface curvature index; NOAA = national oceanic and atmospheric administration; INDOT = Indiana department of transportation; NHS-NIS = national highway system–non-interstates; IS = interstate highways; INDIPave = database for Indiana state; PCR = pavement condition rating; PI = precipitation index; TI = temperature index; PCI = pavement condition index; AC = asphalt concrete pavements; PCC = portland cement concrete pavements; and PA = pavement age.

^aMultiple regions: Indiana, Iowa, Maryland, New Jersey, New York, Tennessee, Arkansas, and Oklahoma in United States; New Brunswick and Prince Edward Island in Canada.

subgrade resilient modulus (Hong and Somo 2001; Abaza 2004), construction quality (Rose et al. 2016), and pavement treatment expenditure (Fettahoglu 2018).

Pavement performance is modeled using deterministic, probabilistic, and artificial intelligence–based approaches. Deterministic models presume a symmetrical scatter around the average value of pavement condition, which means that 50% of the actual values are higher or lower than the predicted values. Deterministic models are categorized into three groups: mechanistic, empirical (regression models), and mechanistic-empirical models (Ferreira and Santos 1999; Khan et al. 2014). Regression models can be either univariate or multivariate, simple or multiple, and linear or nonlinear, based on the data availability and the relationship between the dependent and independent variables. Regression models can comprise either fixed or random parameters. Fixed-parameters (FPs) models assume that the estimable parameters do not change across observations. Abaza (2004) developed fixed-parameters multiple linear and nonlinear regression models to predict pavement condition. Dalla Rosa et al. (2017) developed fixed-parameters nonlinear regression models to predict pavement IRI using data of pavement age and initial IRI (post construction or rehabilitation) for different functional classes of highways. Saeed et al. (2017) estimated fixed-parameters nonlinear models to relate posttreatment pavement performance to the accumulated annual truck traffic loadings and accumulated annual freezing index for different pavement surface and treatment types and highway functional classes.

Conversely, random-parameters (RPs) models consider the estimable parameters as random variables across observations with a specific probability distribution. Random-parameters models help account for the unobserved heterogeneity, i.e., the existence of influential factors that could contribute to the deterioration of pavement performance but are not captured in the existing set of explanatory variables (Anastasopoulos et al. 2016; Sarwar and Anastasopoulos 2016; Nahidi et al. 2017; Sarwar et al. 2019). Random-parameters models avoid erroneous outcomes by accounting for the possibility of bias and inconsistency in the estimation process (Washington et al. 2011), and hence were used extensively in previous studies (Anastasopoulos et al. 2011, 2012; Anastasopoulos and Mannering 2014; Ahmed et al. 2017; Anastasopoulos et al. 2017; Chen et al. 2017, 2019; Fountas et al. 2019; Saeed et al. 2019; Waseem et al. 2019; Yamany and Elwakil 2019).

Probabilistic approaches to pavement performance modeling are more reliable, relevant, and robust than deterministic approaches. Hence, several studies (Amador-Jiménez and Mrawira 2011; Lethanh and Adey 2013; Moreira et al. 2018) developed probabilistic pavement performance models because they duly acknowledge the uncertainties inherently attributed to pavement condition data. These models are recommended for modeling pavement condition mainly because of the uncertain nature of pavement condition, the measurement error, the nonlinearity of pavement deterioration trends, and the heterogeneity resulting from the unobserved influential factors (Madanat et al. 1995; Abaza 2016).

Artificial intelligence techniques, including artificial neural networks and fuzzy logic, have been used to develop pavement performance models (Amin and Amador-Jiménez 2016; Plati et al. 2016; Ziari et al. 2016; Yamany et al. 2019). ANNs models are capable of dealing with the complexity of pavement structure and nonlinearity of measured data (Bianchini and Bandini 2010). Plati et al. (2016) developed an ANNs model to assess and predict pavement strains using pavement thickness and deflection. An ANNs model was integrated with the generalized delta rule (GDR) learning algorithm to estimate pavement PCI while reducing the measurement errors of pavement performance modeling (Amin and Amador-Jiménez 2016). An ANNs model was incorporated with

the group method of data handling (GMDH) to predict pavement IRI (Ziari et al. 2016). Moreover, fuzzy logic was used in conjunction with an ANNs model to predict pavement PSI, producing a robust framework and resulting in accurate findings (Bianchini and Bandini 2010).

Motivation and Problem Statement

The present study hypothesized that individual pavement performance models should be estimated for each state independently for use in the pavement management system due to variation across the states in terms of agency practices, business processes, funding availability, pavement maintenance expenditures, operating environment, and work activity profile, among several other factors. Past studies developed pavement performance models using data from across the states, for instance, the LTPP data from the whole United States, but did not investigate the performance of these models for individual states. Mazari and Rodriguez (2016) developed pavement performance models using data from Indiana, Iowa, Maryland, New Jersey, New York, Tennessee, Arkansas, and Oklahoma in the United States, and New Brunswick and Prince Edward Island in Canada. The relative performance of the models was explored for the entire data set; however, the performance of the models when applied to a single state has yet to be determined. This also was the case in several other studies (Sollazzo et al. 2017; Abdelaziz et al. 2020). Hence, it can be hypothesized that a model that has been optimized to predict pavement performance across different jurisdictions may not be optimal for a single or individual jurisdiction. The present study investigated this hypothesis using LTPP data of in-service Interstate flexible pavements from eight Midwestern states: Indiana, Illinois, Wisconsin, Michigan, Ohio, Minnesota, Iowa, and Missouri.

Another important consideration of this study was the unobserved heterogeneity that arises in data collected across jurisdictions due to variables that can affect pavement performance but typically are not measured or captured in the data set. These variables include the quality of pavement construction materials and workmanship, the type of drainage, and the intensity and frequency of deicing applications during winter season. In the case of unobserved heterogeneity, the influence of explanatory variables is potentially heterogeneous across different pavement segments. In instances in which the parameter estimates are constrained to be constant although they possibly may vary across the observations, the resulting fixed-parameters model could produce biased estimates. To investigate the unobserved heterogeneity, fixed- and random-parameters models were estimated, and their relative statistical superiority was investigated.

Exploratory Data Analysis

Exploratory data analysis starts with retrieving data from a source database, followed by preprocessing (detecting and removing

extreme outliers and leverage points). The variation in the data across the study states then is investigated. These steps are discussed in detail in the ensuing subsections.

Data Collection and Collation

The Long-Term Pavement Performance program, started in 1987, was part of the Strategic Highway Research Program (SHRP), but has been managed by the Office of Infrastructure Research and Development of Federal Highway Administration (FHWA) since 1991. This program monitors and collects pavement performance data for over 2,500 in-service highway test sections throughout North America (FHWA 2018). The LTPP database contains information in seven modules: inventory, maintenance, monitoring, rehabilitation, material testing, traffic, and climate data. The Interstate flexible pavement LTPP data used in this study covered from 1989 to 2016 for Indiana, Illinois, Wisconsin, Michigan, Ohio, Minnesota, Iowa, and Missouri. The initial number of pavement condition observations obtained from each state was 1,159. The data included the response variable represented in pavement roughness, which is measured by the International Roughness Index—widely used as pavement condition indicator (Table 1). The explanatory variables include annual average precipitation (AAP), annual average temperature (AAT), annual average freezing index (AAFI), annual average daily truck traffic (AADTT), equivalent single-axle loads (ESALs), and pavement age in years.

Data Preprocessing

The collected data were preprocessed by detecting and removing the extreme outliers and leverage points. The extreme outliers were the data points with values out of the range of the mean value $\pm 3\sigma$ (Luo 2013; Ahmed et al. 2016). The extreme leverage points were those observations that had leverage higher than the allowable value. The statistical allowable leverage value was set to $2p/n$ (Belsley et al. 1980), where p is the number of parameters and n is the number of observations. After preprocessing the data, the total number of observations was reduced to 966; 10% of the data set, or 96 observations, were randomly sampled to be used for model validation, whereas the remaining 90% of the data set (i.e., 870 observations) was used for model estimation. The description and summary statistics of each variable are presented in Table 2. Some Interstate flexible pavements were either newly constructed or had undergone major rehabilitation recently, and others were as old as 27 years.

Variation Investigation

The data were expected to vary across the states. Figs. 1(a–g) show the variation between the mean values of the variables from each state and the mean values of the variables from all Midwestern states (i.e., the entire data set). Pavements with the best average condition (lower IRI) compared with the mean value were found

Table 2. Descriptive statistics of key study variables

Variable	Description	Mean	Standard deviation	Min	Max
IRI	International roughness index (m/km)	1.28	0.44	0.49	3.08
Pavement age	Years since construction or rehabilitation	9.6	6.6	0.0	27.0
AAP	Annual average precipitation (m)	0.97	0.19	0.62	1.56
AAT	Annual average temperature (°F)	50.6	3.4	42.8	59.7
AAFI	Annual average freezing index (°F days)	788	393	70	1,924
AADTT	Annual average daily truck traffic (trucks)	2,169	802	249	5,115
ESALs	Equivalent single axle loads (18-kip)	1,115	534	122	3,195

in Michigan, whereas the worst average condition was found in Minnesota, with a difference between the two states of approximately 0.789 m/km. [Fig. 1(a)]. The youngest pavements were found in Minnesota, and the oldest were found in Ohio [Fig. 1(b)].

The state of Indiana was found to have higher annual precipitation levels, higher average temperatures, and lower AAFI values, whereas the state of Minnesota was found to have lower annual precipitation levels, lower average temperatures, and higher AAFI

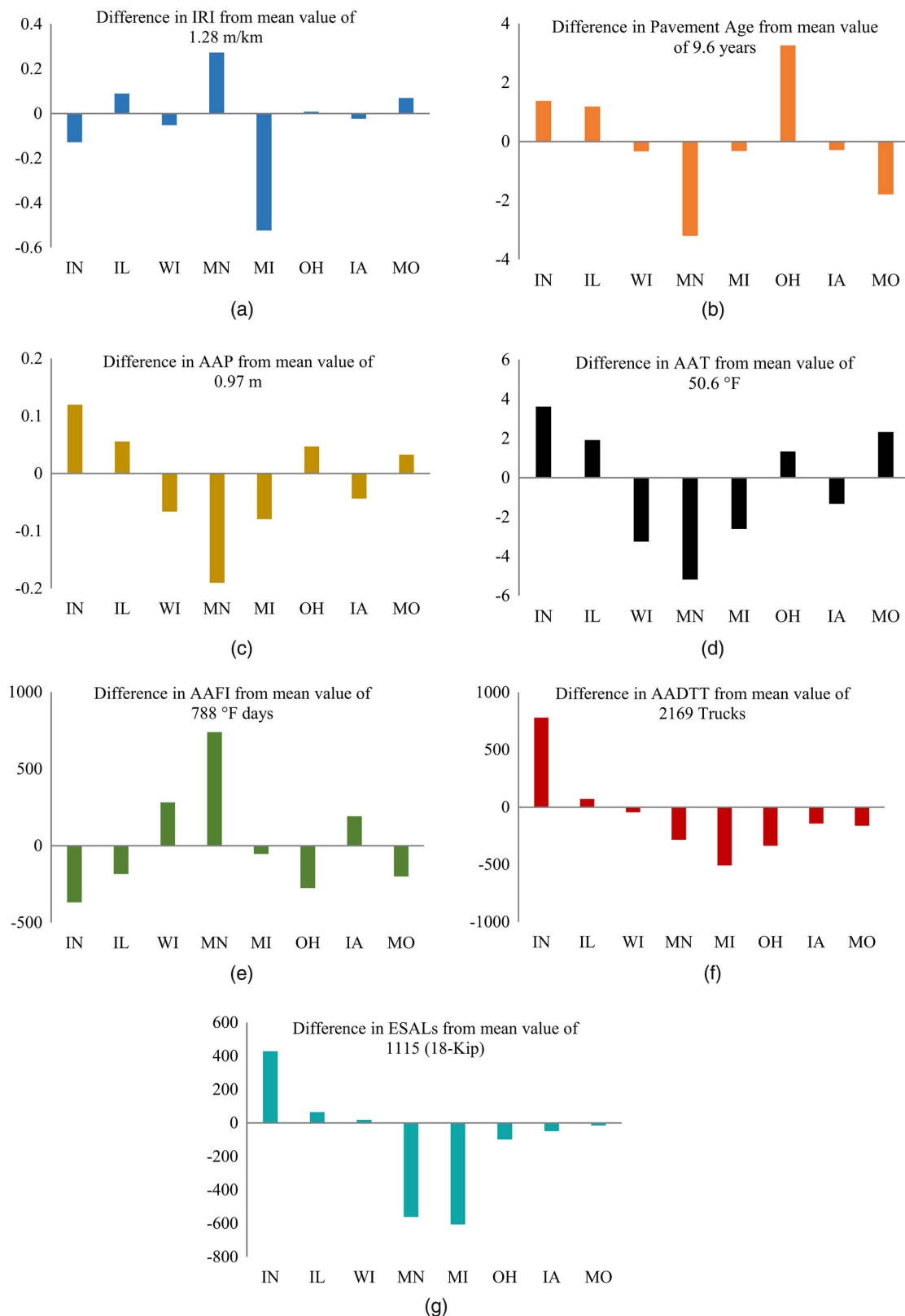


Fig. 1. Variation of LTPP data of interstate flexible pavements across Midwestern states: (a) IRI; (b) pavement age; (c) AAP; (d) AAT; (e) AAFI; (f) AADTT; and (g) ESALs.

values [Figs. 1(c–e)]. With respect to traffic loading, the state of Indiana was found to have the largest annual traffic volume, whereas the states of Minnesota and Missouri were found to have a lower annual traffic volume in terms of AADTT and ESALs [Figs. 1(f and g)]. Fig. 1 clearly indicates a variation in the study variables across the states.

Development of Models and Discussion of Results

The methodology and framework of the current paper are shown in Fig. 2. Statistical regression and ANNs models were developed to estimate and predict the IRI of the Interstate flexible pavements in the Midwestern states using the explanatory variables discussed in the previous section. This section elaborates the methodological framework, the models' development process, and the results and related discussion.

Fixed-Parameters Regression Model

A fixed-parameters regression model was estimated to facilitate later evaluation of the random-parameters model. The FPs model is recommended when the data variability is insignificant. Due to the continuous nature of the response variable (IRI), a standard regression model that uses the ordinary least-squares (OLS) method was estimated, such that

$$Y_i = \beta_0 + \beta_i X_i + \varepsilon_i \quad (1)$$

where Y_i = response variable (pavement roughness, i.e., IRI); β_0 = constant term; β_i = vector of estimated parameters; X_i = vector of explanatory variables; and ε_i = normally distributed disturbance term.

Several explanatory variables were created to capture the interaction between the variables in Table 2. Examples of the created variables are accumulated annual average precipitation (AAP $\times$ pavement age), accumulated annual average freezing index (AAFI $\times$ pavement age), and accumulated annual average daily truck traffic (AADTT $\times$ pavement age). Mallow's C_p statistic and stepwise selection techniques were used to choose the statistically significant independent variables to be included in the

proposed models at a confidence level of 95% (Draper and Smith 1998). Although past studies (Table 1) reported several variables that affect pavement condition, the current statistical analysis found that only pavement age and AAFI were statistically significant in the final model.

After estimating the FPs model, fundamental diagnostics were used to assess the model's validity by checking the linear regression assumptions (normality, linearity, and constant variance). The residual plots were used to test for the linearity and constant variance/homoscedasticity, whereas the histogram plot was used to test for the normality. Fig. 3(a) illustrates that the residuals of the IRI across pavement age and AAFI had no specific pattern, which implies that these regressors had a linear relationship with the IRI. Moreover, the residuals of IRI by pavement age and AAFI were nearly symmetrically scattered around the zero-residual line, indicating the homoscedasticity or constant variance. Fig. 3(b) shows the histogram of the IRI residuals, which indicates that the IRI residuals were normally distributed. Additionally, a Shapiro–Wilk test was conducted to check the normality assumption (Royston 1982). The W -statistic was found to have a value of 0.954 with a p -value less than 0.0001, which indicates that the distribution of the IRI residuals was statistically normal at a 95% confidence level. In summary, none of the linear regression assumptions was found to be violated.

To test the possibility of autocorrelation in the residuals due to the time-series nature of pavement condition data, the Durbin–Watson (DW) statistic was estimated. If the DW-statistic value is significantly different from 2.0, the autocorrelation is deemed statistically significant. The DW statistic was found to be significantly different from 2.0 at a 95% confidence level, indicating the presence of autocorrelation between the study observations. As such, the Prais–Winsten algorithm was implemented to correct for this autocorrelation/serial correlation (Greene 2011). After implementing this correction, the value of the DW statistic was 2.02, which is not statistically different from 2.0 at a 95% confidence level.

The final best statistical model was selected based on the values of the coefficient of determination (R^2) and the RMS error (RMSE). The multiple linear regression model in Eq. (2) and Table 3 was found to best fit the data

$$\text{IRI} = 0.8417 + 0.0001\text{AAFI} + 0.0394(\text{pavement age}) \quad (2)$$

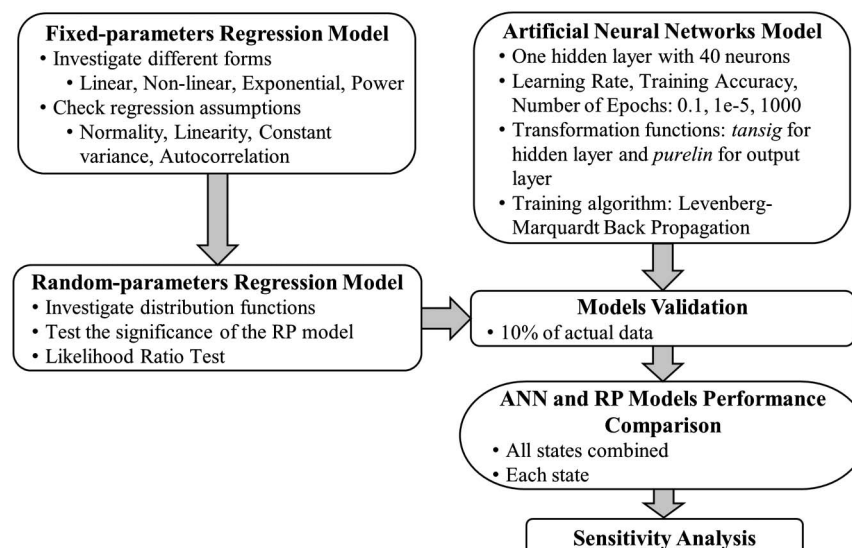


Fig. 2. Research framework.

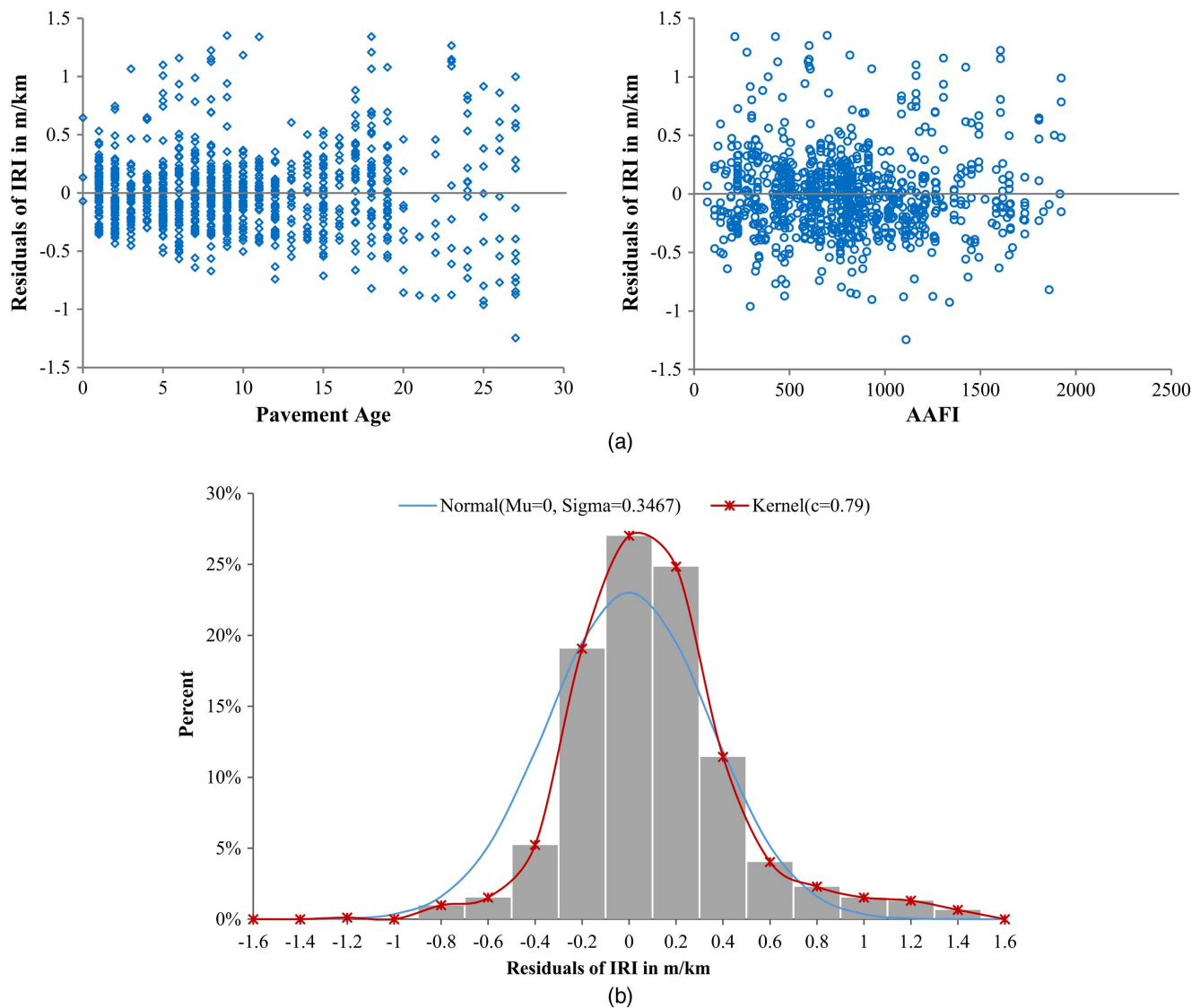


Fig. 3. Regression diagnostic plots: (a) residual plots for IRI across pavement age and AAFI; and (b) histogram plot of IRI residuals.

Table 3. Estimation results of fixed- and random-parameters models

Variable	Fixed-parameters model		Random-parameters model	
	Parameter estimate	<i>t</i> -statistic	Parameter estimate	<i>t</i> -statistic
Constant	0.8417	27.17	0.8544 (0.143)	49.75 (24.54)
AAFI	0.0001	3.65	0.000085 (0.000163)	5.56 (24.21)
Pavement age	0.0394	21.58	0.0404 (0.0186)	43.05 (37.94)
Log-likelihood at zero, $LL(\beta_R)$	-3,956.78		-3,956.78	
Log-likelihood at convergence, $LL(\beta_U)$	-3,780.18		-3,715.25	
Number of observations	870			

Note: The standard deviation for random parameters is presented in parentheses.

The adjusted R^2 and RMSE of the multiple linear regression model were found to be 0.364 and 0.349 m/km, respectively. The value of the adjusted R^2 indicates that 36.4% of the variation in the condition of the Interstate flexible pavements across the eight Midwestern states was explained by the FPs model. The value of the RMSE means that the FPs model can be used to predict the IRI of the Interstate flexible pavements across the eight Midwestern states with a mean error of 0.349 m/km.

Random-Parameters Regression Model

In fixed-parameters regression modeling, the explanatory variables are assumed to have the same effect across all observations in the data set. However, the effect of the explanatory variables most probably is different across individual observations due to the heterogeneous operating environment. Data collected from different locations contribute considerably to the data variation due to its

significant spatial effects (Anastasopoulos et al. 2010). In this study, the data were from eight different states with different pavement maintenance activity profiles and agency business processes; therefore, an unobserved heterogeneity was highly likely. To account for the possibility of unobserved heterogeneity, a random-parameters regression model was proposed to estimate pavement condition. The RPs regression models estimate the random parameters using the maximum-likelihood estimation (MLE) method, which also is employed to account for the autocorrelation/serial correlation (Greene 2011). The estimated random parameter β_r is defined as

$$\beta_r = \beta_i + \mu_r \quad (3)$$

where μ_r is a disturbance term with mean = 0 and standard deviation σ . If this term is statistically insignificant, i.e., if its standard deviation is not significantly different from zero, the term reduces to a fixed parameter β_i .

Table 2 contains the explanatory variables that were available for the empirical exploration in this study. Nevertheless, other, unobserved factors, such as pavement layer thickness, type of base course material, and construction quality also can influence pavement condition, and were accounted for by estimating the RPs regression model. The RPs regression models assume that the estimable parameters are random variables with probability distribution functions (normal, exponential, and so forth). Various possible distribution functions were investigated for each of the estimable parameters, and the statistical significance of their standard deviations were tested using the likelihood-ratio test. The normal distribution was found to be statistically superior to its counterparts. The estimation results of the best RPs model are presented in Eq. (4) and Table 3

$$\text{IRI} = 0.8544 + 0.000085\text{AAFI} + 0.0404(\text{pavement age}) \quad (4)$$

The suitability of the RPs model was assessed by conducting the likelihood-ratio test and calculating the computed chi-square (χ^2) (Washington et al. 2011)

$$\chi^2 = -2[\text{LL}(\beta_R) - \text{LL}(\beta_U)] \quad (5)$$

where $\text{LL}(\beta_R)$ = log-likelihood at convergence of restricted model; and $\text{LL}(\beta_U)$ = log-likelihood at convergence of unrestricted model. The $\text{LL}(\beta_R)$ and $\text{LL}(\beta_U)$ were found to be $-3,956.87$ and $-3,715.25$, respectively. Thus, the computed χ^2 was 483.24. The critical χ^2 at a 95% confidence level and six degrees of freedom (the difference between the number of parameters in the restricted and the unrestricted models) was 12.59 (Washington et al. 2011), which was less than the computed χ^2 . Therefore, the RPs model was statistically significant with a confidence level of 95%. Furthermore, the t -statistic was calculated to examine the statistical significance of the estimated random parameters and their standard

deviations (the values in parentheses in Table 3). The calculated t -statistic indicates the statistical significance of the estimated parameters and their standard deviations at a 95% confidence level. The statistical significance of the standard deviations of the estimated parameters ascertained that the effect of the independent variables (pavement age and AAFI) on pavement condition varied across observations.

The constant term of the FPs model reported in Table 3 indicates that the expected condition of newly constructed pavements was 0.841 m/km. However, the constant term of the RPs model implies that the expected condition of newly constructed pavements was 0.855 ± 0.286 m/km (mean value ± 2 standard deviations) at a 95% probability, which is more realistic given the actual data. In addition, the results of the RPs model showed that the change in the AAFI and pavement age by one unit influenced pavement condition by 0.000085 ± 0.000326 m/km and 0.040 ± 0.0372 m/km, respectively, with a likelihood of 95%. This in turn accounted for the effect of the unconsidered factors such as traffic loading and pavement maintenance and rehabilitation (M&R) that could cause degradation in or improvement of pavement condition.

A likelihood-ratio test was carried out to investigate and establish the relative statistical superiority of the competing random- and fixed-parameters models as follows:

$$\chi^2 = -2[\text{LL}(\beta_{FP}) - \text{LL}(\beta_{RP})] \quad (6)$$

where $\text{LL}(\beta_{FP})$ = log-likelihood at convergence of fixed-parameters model; and $\text{LL}(\beta_{RP})$ = log-likelihood at convergence of random-parameters model. The $\text{LL}(\beta_{FP})$ and $\text{LL}(\beta_{RP})$ were found to be $-3,780.18$ and $-3,715.25$, respectively. Thus, the computed χ^2 was 129.86. The critical χ^2 at a 95% confidence level and four degrees of freedom (the difference between the number of parameters in the fixed- and random-parameters models) was 9.49 (Washington et al. 2011), which was less than the computed χ^2 . This ascertained that the developed random-parameters model outperformed its fixed-parameters counterpart. Hence, this proves one of the two main hypotheses of this study that there is an unobserved heterogeneity and the influence of the explanatory variables on the response variable is different across different pavement segments. Due to its statistical superiority, the RPs model was used for the subsequent analysis.

Artificial Neural Network Model

The architecture of the proposed ANNs model for estimating and predicting pavement roughness (IRI) is shown in Fig. 4. The ANNs architecture comprises input, a hidden layer, an output layer, and output. The input represents the explanatory variables and consists of two neurons: one for the pavement age and the other for the AAFI. The output contains one neuron representing the IRI response variable.

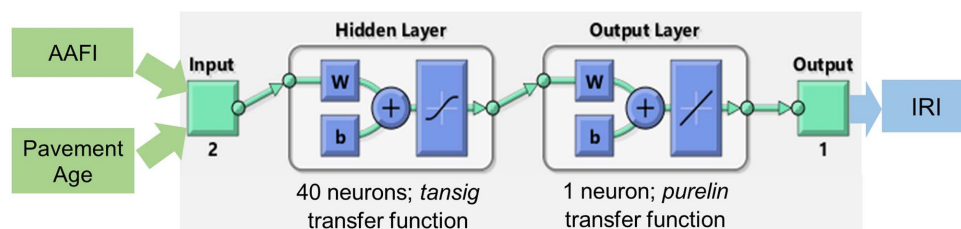


Fig. 4. ANNs model architecture implemented in this study.

The ANNs model was designed through identifying the following parameters: number of hidden layers, number of neurons in each hidden layer, learning rate, training accuracy, number of epochs, transfer functions for the hidden and output layers, and training algorithm. Although the number of hidden layers in the ANNs models was not clearly defined in the literature, most prior studies assumed one or two hidden layers. Zhang and Gupta (2000) and Choi et al. (2004) concluded that one hidden layer was sufficient to capture all variations and features of the data used in their ANNs models. Ziari et al. (2016) reported that an ANNs model with one hidden layer can predict pavement condition with accuracy greater than that of models with two or three hidden layers. Based on the guidance from prior studies, one hidden layer was assumed for the proposed ANNs model for the current study.

Prior researchers identified the number of neurons in the hidden layers using trial and error or heuristic methods (Ziari et al. 2016). The present paper estimated the number of neurons in the hidden layer by trial and error. Different numbers of neurons were assumed, the R^2 of each trial was observed, and the number of neurons associated with the highest R^2 was selected. The highest R^2 , 0.66, was obtained when the number of neurons was 40; therefore, the number of neurons in the hidden layer was assumed to be 40 for the ANNs design. Hence, the neural network structure can be read as 2-40-1, indicating 2 neurons in the input layer, 40 neurons in the hidden layer, and 1 neuron in the output layer (Fig. 4).

The smaller the learning rate and training accuracy and the larger the number of epochs, the more likely it is that the global optimal point can be located. As is typical practice in developing ANNs models, the learning rate, training accuracy, and number of epochs were assumed to be 0.1, 1×10^{-5} , and 1,000, respectively. Different transfer functions such as tangent sigmoid, logistic sigmoid, and linear were assumed for the hidden and output layers. The choice of the optimal transfer function was based on the value of R^2 of the estimated model. The tangent sigmoid (tansig) function was chosen for the hidden layer, whereas the linear (purelin) function was chosen for the output layer. Several studies, such as Sobhani et al. (2010) and Ziari et al. (2016), adopted the Levenberg–Marquardt back-propagation (LMBP) algorithm to train their ANNs models, and found it computationally superior to other algorithms such as standard back-propagation. As such, the LMBP algorithm was employed in the present study to train an ANNs model for pavement roughness prediction.

To avoid the bias resulting from the large difference between observations (maximum and minimum values), data were normalized prior to the ANNs modeling. Data normalization also facilitated neural network learning, which in turn improved the performance of the designed ANNs model. Each value of the response and explanatory variables ranged from 0.1 to 0.9. Eq. (7) shows the calculation of the normalized value $x_{i(n)}$, where x_i , $i_{\max}$, and $i_{\min}$ are the measured, minimum and maximum values (Ghafoori et al. 2013)

$$x_{i(n)} = 0.1 + 0.8 \times \frac{x_i - i_{\min}}{i_{\max} - i_{\min}} \quad (7)$$

One major shortcoming of ANNs models is that their framework provides limited insight into the approximated function, and their results are difficult to interpret (García de Soto et al. 2018). Due to this shortcoming, past research criticized ANNs models for using a black-box technique (Hegazy et al. 1994; Boussabaine 1996). Accordingly, sensitivity analysis has been conducted to explain the influence of key predictors on their respective response variables (Chang 2005; Delen et al. 2006). Therefore, a sensitivity analysis was carried out to analyze the impact of the explanatory variables on the deterioration of pavement performance.

Model Validation

To evaluate the relative performance of the competing statistical regression and ANNs models, a model validation process is useful for assessing the models' capabilities to predict unseen or out-of-sample data. To test the validity of the developed models, 10% of the data (96 observations) were set aside and used to predict IRI values; then the predicted values were compared with the actual IRI values. The validity and performance of statistical models typically are assessed using different validation and performance measures. This study used the mean absolute deviation (MAD), mean absolute percentage error (MAPE), mean square error (MSE), RMSE and R^2 to measure the validity and compare the performance of the statistical regression and ANNs models, which were calculated as follows:

$$\text{MAPE} = \frac{1}{N} \sum_i \left(\frac{|\text{IRI}_{i,\text{observed}} - \text{IRI}_{i,\text{predicted}}|}{\text{IRI}_{i,\text{observed}}} \right) \times 100 \quad (8)$$

$$\text{MSE} = \frac{1}{N} \sum_i (\text{IRI}_{i,\text{observed}} - \text{IRI}_{i,\text{predicted}})^2 \quad (9)$$

$$R^2 = 1 - \frac{\sum_i (\text{IRI}_{i,\text{observed}} - \text{IRI}_{i,\text{predicted}})^2}{\sum_i (\text{IRI}_{i,\text{observed}} - \overline{\text{IRI}})^2} \quad (10)$$

Of the statistical regression models, the random-parameters model was chosen for the relative comparison because of its superiority to the fixed-parameters model (Table 3). The calculated validation measures for the RPs and ANNs models are summarized in Table 4. An R^2 of 0.4 to 0.6 is an indication of a moderate correlation, whereas R^2 between 0.6 and 0.8 signifies a strong correlation (Rajagopal 2006). Therefore, both the RPs and ANNs models had good overall performance. However, the ANNs model outperformed the RPs regression, with 26.14%, 26.14%, 20.93%, 11.08%, and 47.92% improvement in the values of MAD, MAPE, MSE, RMSE, and R^2 , respectively. Hence, the ANNs model was identified as the best model to predict the IRI of Interstate flexible pavements using data from all states combined.

The RMSE values of the developed models are presented in relation to those reported in similar past studies in Table 5. Although no direct comparison is possible because different data sets were

Table 4. Comparison of models' predictability for entire data set

Performance measure	RP	ANN	Prediction improvement (%)
MAD	0.385 (m/km)	0.285 (m/km)	26.12
MAPE	25.44 (%)	18.79 (%)	26.14
MSE	0.0894	0.0707	20.93
RMSE	0.2991 (m/km)	0.2659 (m/km)	11.08
R^2	0.48	0.71	47.92

Table 5. RMSE values of present and past models

IRI prediction models	Modeling techniques	RMSE (m/km)
Present study	ANNs	0.2659
	RP regression	0.2991
La Torre et al. (1998)	ANNs	0.6399
Mazari and Rodriguez (2016)	Hybrid ANNs and GEP	0.0489
Ziari et al. (2016)	ANNs	0.3499
Dalla Rosa et al. (2017)	Nonlinear FP regression	0.3156

used for these studies, it does give an idea of the extent of the models' prediction errors.

Performance Comparison of ANNs and RPs Models for Individual States

The empirical data from each of the eight Midwestern states were individually fed into the developed RPs and ANNs models to assess their performance for each individual state. To explore the performance of the developed models, scatter plots of the actual and the predicted IRI using RPs and ANNs models were developed [Figs. 5(a–h)]. This graphical analysis readily illustrates a wide range of aspects of the relationship between the developed models and the actual observations. Figs. 5(a–h) show the fit of the ANNs and RPs models to the actual data from each state.

The ANNs model fit better to the data from five states, Illinois, Wisconsin, Minnesota, Iowa, and Missouri, with $R^2 = 0.48, 0.72, 0.57, 0.71,$ and 0.69 , respectively (Figs. 5 and 6). Conversely, the RPs model fit better to the data from three states, Indiana, Michigan, and Ohio, with $R^2 = 0.51, 0.42,$ and 0.74 , respectively. It is hard to generalize the performance capability of these models when they are evaluated for individual states (Fig. 6). This supports the hypothesis that a variation exists in pavement-related information across the states. As a result, state highway agencies and pavement asset managers should rely on models using pavement data from their own jurisdictions rather than those using data from the whole country or other jurisdictions that are significantly different from theirs.

The significance of spatial and temporal variation in the data and subsequently the ability of the comprehensive model to transfer across states can be assessed using a transferability/likelihood-ratio test (Washington et al. 2011). The transferability test requires the estimation of separate models for each state. However, a few states did not have the prerequisite number of observations needed to properly estimate the model. Therefore, the transferability of the comprehensive model across all states could not be confirmed using the transferability test approach. If subsequent analysis suggests that the comprehensive model is nontransferable, separate models must be estimated for each state. In that case, a comparison of the forecasting accuracy should be based on the predictors of each state's model. Furthermore, to facilitate a fair comparative analysis, the ANNs models should be trained and calibrated for each state, although a few states had a limited number of observations. The comparison of the forecasting accuracy of the RPs and ANNs models in the present study possibly could favor the ANNs model for the following reasons:

1. The RPs model could suffer from omitted-variable bias. Only two variables were found to be statistically significant in the present model, unlike previous research that found additional factors that influence pavement performance. The prospect that omitted factors are present was reinforced by the fact that both explanatory variables produced random parameters. This limitation of the data set has an influence on the explanatory power and statistical performance of the model.
2. In this study, the ANNs approach was acknowledged for its high predictive performance and limited explanatory power. The forecasting accuracy of statistical methods is highly reliant on how well-specified the models are. Therefore, evaluating the performance based on the predictive ability of the ANNs and the RPs models possibly may be biased given the nature and size of the data set and the resulting possibility of misspecification of the RPs model.

Sensitivity Analysis

The sensitivity of the pavement roughness (i.e., IRI) to changes in the independent variables pavement age and AAFI was evaluated by varying the value of one independent variable from its minimum to its maximum value while holding the other independent variable at its mean value. These new sets of input then were fed into the developed ANNs and RPs models to predict pavement IRI. Figs. 7(a and b) show the sensitivity plots for IRI versus pavement age and IRI versus AAFI, respectively, using the developed ANNs model and the developed RPs model at the 25, 50, and 75 percentiles. These plots indicate that pavement IRI is more sensitive to pavement age than to AAFI.

Conclusions

This study was motivated by the hypothesis that estimates of pavement performance optimized using cross-jurisdictional (state) data may not be the most optimal estimate for a particular state. This is due to the variation across states in terms of agency practices, business processes, funding availability, pavement maintenance expenditures, operating environment, and work activity profile, among several other factors. Therefore, departments of transportation should rely on pavement performance models using data from their own jurisdiction and not from jurisdictions that are significantly different from their own. Furthermore, regression models of pavement performance should account for the unobserved heterogeneity arising from the existence of variables that could affect pavement performance but are not captured in the study data set. To investigate these hypotheses, this study used the LTPP data of Interstate flexible pavements from eight Midwestern states (Indiana, Illinois, Wisconsin, Michigan, Ohio, Minnesota, Iowa, and Missouri) from 1989 to 2016.

To test the study hypotheses, both fixed- and random-parameters regression models were estimated, and their relative performance was investigated. The RPs model accounting for the unobserved heterogeneity was found to perform better than its counterpart. The independent variables annual average freezing index and pavement age were determined to have random parameters that followed a normal distribution. Through the RPs model, this study duly acknowledges the heterogeneous effects of independent variables across the individual observations in the data set from eight different states.

In addition, this paper presented an ANNs model, widely used to characterize pavement performance. The relative performance of the ANNs and RPs models was evaluated for the entire data set using the statistical performance measures R^2 , MSE, RMSE, MAD, and MAPE. The ANNs model improved prediction performance compared with the RPs model of 47.92%, 20.93%, 11.08%, 26.14%, and 26.14%, respectively, for R^2 , MSE, RMSE, MAD, and MAPE. However, when the performance of these two models was evaluated for each state individually, it was hard to determine what caused one to outperform the other. In three of the eight states, the RPs model was found to outperform the ANNs model. This favors the hypothesis that pavement managers should not rely on models that are developed using data from other jurisdictions significantly different from their own; instead, the models' performance should be evaluated for each individual state.

Although the ANNs model generally had higher predictive power with better forecasting accuracy, the RPs regression model is easier to interpret and duly accounts for unobserved factors responsible for pavement deterioration that are not captured in the data set. Moreover, unlike the ANNs model, the sensitivity of RPs models could provide clearer insights into the influence

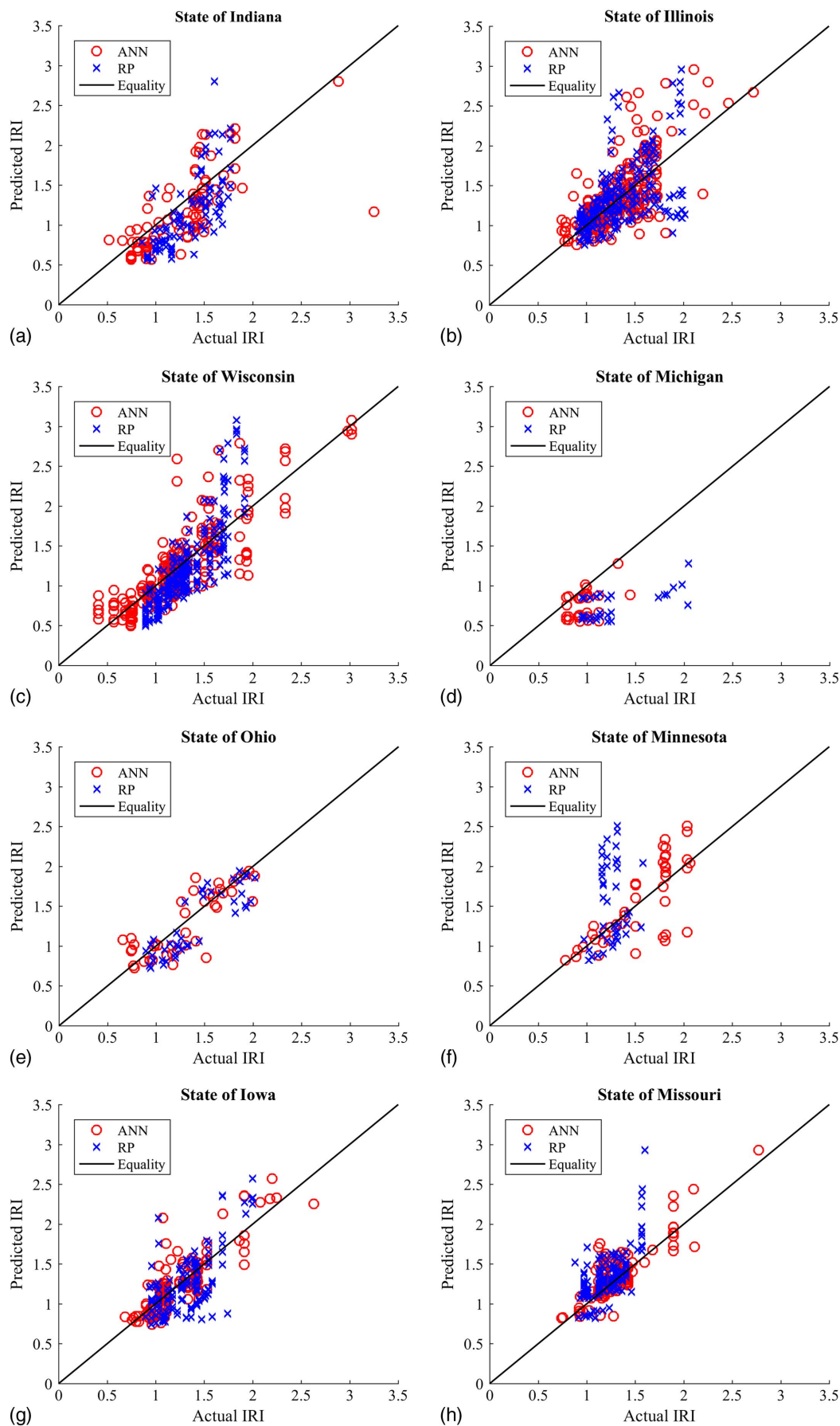


Fig. 5. Fitness of ANNs and RPs models to data from each Midwestern state: (a) Indiana; (b) Illinois; (c) Wisconsin; (d) Michigan; (e) Ohio; (f) Minnesota; (g) Iowa; and (h) Missouri.

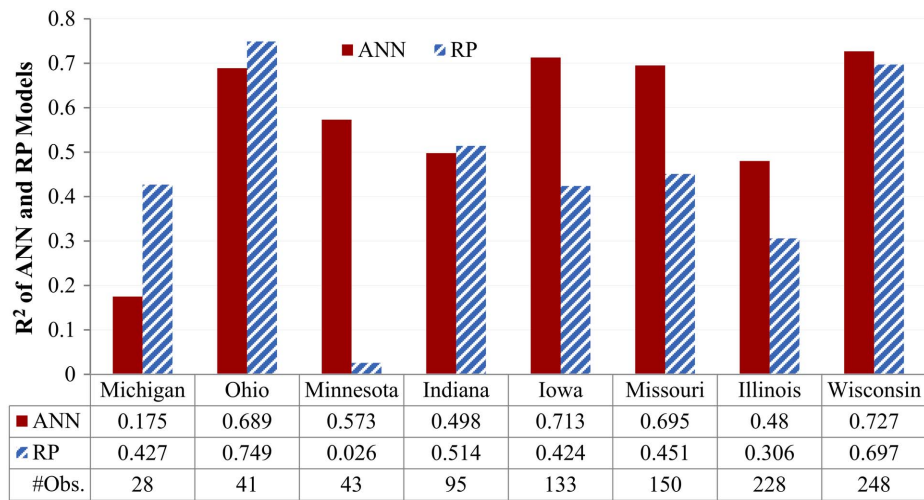


Fig. 6. Change in R^2 of ANNs and RPs models across the Midwestern states.

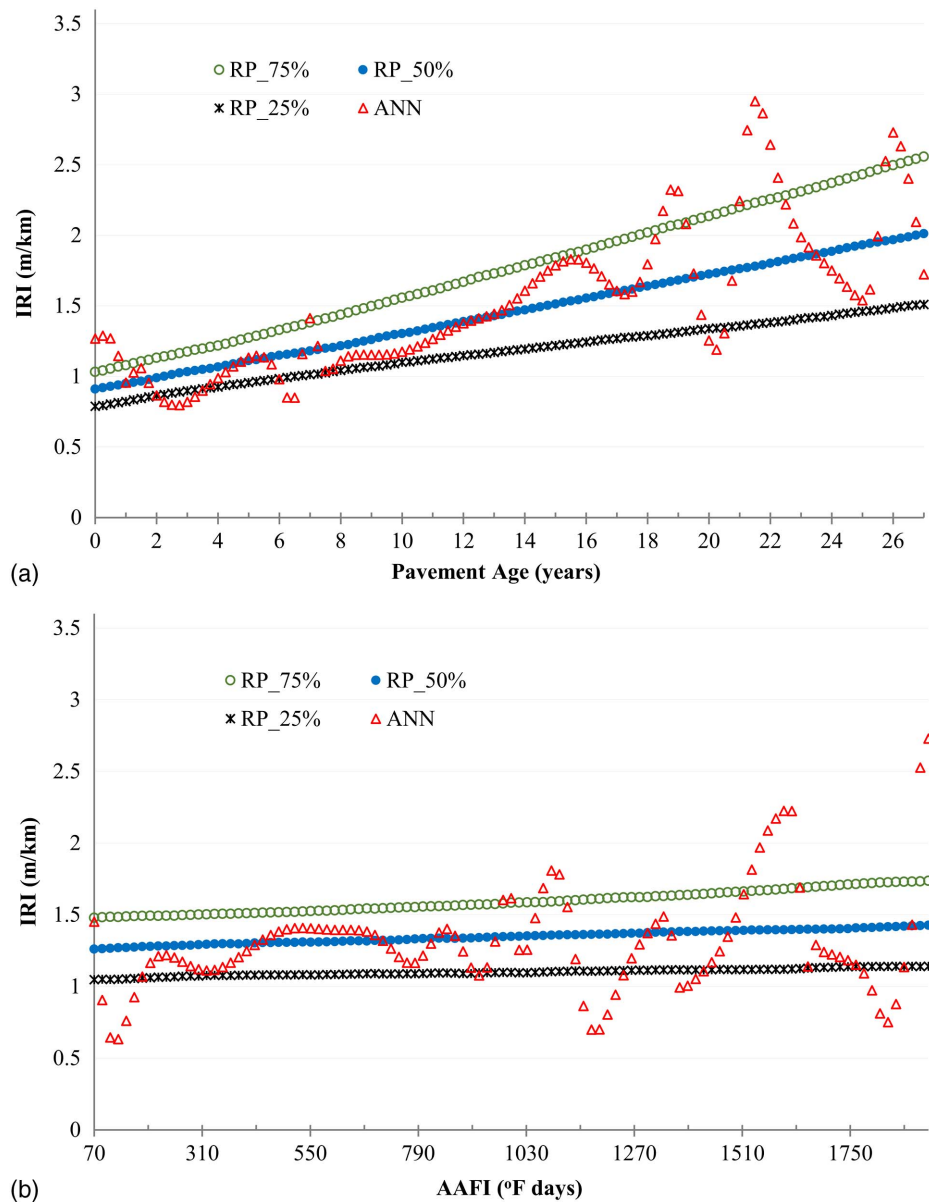


Fig. 7. Sensitivity of IRI to explanatory variables: (a) pavement age; and (b) AAFI.

of explanatory variables on pavement performance. From the signs of independent variables in the RPs model, it was easier to establish that the pavement age and annual average freezing index have negative impacts on pavement performance. The increase in the values of these factors causes an increase in pavement IRI, and hence accelerates pavement deterioration. In contrast, the ANNs model identified the best predictors but did not readily reveal the nature of their influence on the response variable. Unlike the RPs model, it is difficult to make a convenient interpretation of the results of the ANNs model.

Furthermore, a sensitivity analysis was carried out to ascertain the influence of the significant explanatory variables on pavement condition. The sensitivity plots indicated that the effect of increasing one unit of pavement age is greater than that of the AAFI on pavement condition. Despite the contribution of the present paper to the body of knowledge and practice, multiple limitations exist. Only two variables were found to be statistically significant in the developed models, which possibly may cause an omitted variable bias. Thus, it is recommended that this problem be revisited with a larger sample size and a data set containing information on pavement surface material, layer thickness, type of base course material, topography, construction quality, drainage, and type and effectiveness of maintenance and rehabilitation interventions. Because the funding and pavement work activity profile may vary across different states, it is recommended that future studies account for the effect of these and similar other variations (in agency practices and business processes) on pavement performance. Further research studies could also be conducted for other pavement surface types and road functional classes. Moreover, future research may examine the spatial transferability of heterogeneous pavement performance models when sufficient data are available.

Data Availability Statement

Some or all data, models, or code generated or used during the study are available from the corresponding author by request [code only; for data, please refer to the LTTP InfoPave (<https://infopave.fhwa.dot.gov/Data/DataSelection>)]; models are presented in the paper.

References

- Abaza, K. A. 2004. "Deterministic performance prediction model for rehabilitation and management of flexible pavement." *Int. J. Pavement Eng.* 5 (2): 111–121. <https://doi.org/10.1080/10298430412331286977>.
- Abaza, K. A. 2016. "Back-calculation of transition probabilities for Markovian-based pavement performance prediction models." *Int. J. Pavement Eng.* 17 (3): 253–264. <https://doi.org/10.1080/10298436.2014.993185>.
- Abdelaziz, N., R. T. Abd El-Hakim, S. M. El-Badawy, and H. A. Afify. 2020. "International roughness index prediction model for flexible pavements." *Int. J. Pavement Eng.* 21 (1): 88–99. <https://doi.org/10.1080/10298436.2018.1441414>.
- Ahmed, A., T. U. Saeed, and S. Labi. 2016. "Estimation of rest periods for newly constructed/reconstructed pavements." *Transport* 31 (2): 183–191. <https://doi.org/10.3846/16484142.2016.1193050>.
- Ahmed, A., T. U. Saeed, J. Murillo-Hoyos, and S. Labi. 2017. "Pavement repair marginal costs: Accounting for heterogeneity using random-parameters regression." *J. Infrastruct. Syst.* 23 (4): 04017012. [https://doi.org/10.1061/\(ASCE\)IS.1943-555X.0000367](https://doi.org/10.1061/(ASCE)IS.1943-555X.0000367).
- Aleadelat, W., P. Saha, and K. Ksaibati. 2018. "Development of serviceability prediction model for county paved roads." *Int. J. Pavement Eng.* 19 (6): 526–533. <https://doi.org/10.1080/10298436.2016.1176167>.
- Al-Suleiman, T. I., and A. M. Shiyab. 2003. "Prediction of pavement remaining service life using roughness data—Case study in Dubai." *Int. J. Pavement Eng.* 4 (2): 121–129. <https://doi.org/10.1080/10298430310001634834>.
- Amador-Jiménez, L. E., and D. Mrawira. 2011. "Reliability-based initial pavement performance deterioration modelling." *Int. J. Pavement Eng.* 12 (2): 177–186. <https://doi.org/10.1080/10298436.2010.535538>.
- Amin, S. R., and L. E. Amador-Jiménez. 2016. "Backpropagation neural network to estimate pavement performance: Dealing with measurement errors." *Road Mater. Pavement Des.* 18 (5): 1218–1238. <https://doi.org/10.1080/14680629.2016.1202129>.
- Anastasopoulos, P. C., R. J. Florax, S. Labi, and M. G. Karlaftis. 2010. "Contracting in highway maintenance and rehabilitation: Are spatial effects important?" *Transp. Res. Part A: Policy Pract.* 44 (3): 136–146. <https://doi.org/10.1016/j.tra.2009.12.002>.
- Anastasopoulos, P. C., G. Fountas, M. T. Sarwar, M. G. Karlaftis, and A. W. Sadek. 2017. "Transport habits of travelers using new energy type modes: A random parameters hazard-based approach of travel distance." *Transp. Res. Part C: Emerging Technol.* 77 (Apr): 516–528. <https://doi.org/10.1016/j.trc.2017.01.017>.
- Anastasopoulos, P. C., S. Labi, M. G. Karlaftis, and F. L. Mannering. 2011. "Exploratory state-level empirical assessment of pavement performance." *J. Infrastruct. Syst.* 17 (4): 200–215. [https://doi.org/10.1061/\(ASCE\)IS.1943-555X.0000057](https://doi.org/10.1061/(ASCE)IS.1943-555X.0000057).
- Anastasopoulos, P. C., and F. L. Mannering. 2014. "Analysis of pavement overlay and replacement performance using random parameters hazard-based duration models." *J. Infrastruct. Syst.* 21 (1): 04014024. [https://doi.org/10.1061/\(ASCE\)IS.1943-555X.0000208](https://doi.org/10.1061/(ASCE)IS.1943-555X.0000208).
- Anastasopoulos, P. C., F. L. Mannering, and J. E. Haddock. 2012. "Random parameters seemingly unrelated equations approach to the postrehabilitation performance of pavements." *J. Infrastruct. Syst.* 18 (3): 176–182. [https://doi.org/10.1061/\(ASCE\)IS.1943-555X.0000096](https://doi.org/10.1061/(ASCE)IS.1943-555X.0000096).
- Anastasopoulos, P. C., M. T. Sarwar, and V. N. Shankar. 2016. "Safety-oriented pavement performance thresholds: Accounting for unobserved heterogeneity in a multi-objective optimization and goal programming approach." *Anal. Methods Accid. Res.* 12 (Dec): 35–47. <https://doi.org/10.1016/j.amar.2016.10.001>.
- Bektas, F., O. G. Smadi, and M. Al-Zoubi. 2014. *Pavement management performance modeling: Evaluating the existing PCI equations*. InTrans Project Rep. Ames, IA: Institute for Transportation, Iowa State Univ.
- Belsley, D. A., E. Kuh, and R. E. Welsch. 1980. *Regression diagnostics: Identifying influential data and sources of collinearity*. Hoboken, NJ: Wiley.
- Bianchini, A., and P. Bandini. 2010. "Prediction of pavement performance through neuro-fuzzy reasoning." *Comput.-Aided Civ. Infrastruct. Eng.* 25 (1): 39–54. <https://doi.org/10.1111/j.1467-8667.2009.00615.x>.
- Boussabaine, A. H. 1996. "The use of artificial neural networks in construction management: A review." *Constr. Manage. Econ.* 14 (5): 427–436. <https://doi.org/10.1080/014461996373296>.
- Chang, L. Y. 2005. "Analysis of freeway accident frequencies: Negative binomial regression against artificial neural network." *Saf. Sci.* 43 (8): 541–557. <https://doi.org/10.1016/j.ssci.2005.04.004>.
- Chen, S., T. U. Saeed, S. D. Alqadhi, and S. Labi. 2019. "Safety impacts of pavement surface roughness at two-lane and multi-lane highways: Accounting for heterogeneity and seemingly unrelated correlation across crash severities." *Transportmetrica A: Transp. Sci.* 15 (1): 18–33. <https://doi.org/10.1080/23249935.2017.1378281>.
- Chen, S., T. U. Saeed, and S. Labi. 2017. "Impact of road-surface condition on rural highway safety: A multivariate random parameters negative binomial approach." *Anal. Methods Accid. Res.* 16 (Dec): 75–89. <https://doi.org/10.1016/j.amar.2017.09.001>.
- Choi, J., T. M. Adams, and H. U. Bahia. 2004. "Pavement roughness modeling using back-propagation neural networks." *Comput.-Aided Civ. Infrastruct. Eng.* 19 (4): 295–303. <https://doi.org/10.1111/j.1467-8667.2004.00356.x>.
- Dalla Rosa, F., L. Liu, and N. G. Gharaibeh. 2017. "IRI prediction model for use in network-level pavement management systems." *J. Transp. Eng. Part B: Pavements* 143 (1): 04017001. <https://doi.org/10.1061/JPEODX.0000003>.

- David, D., G. Fountas, M. T. Sarwar, U. Eker, S. Akpinar, and P. C. Anastasopoulos. 2017. *Effectiveness of various public private partnership pavement rehabilitation treatments: A big data informatics survival analysis of pavement service life*. TransInfo Research Project Final Rep. Buffalo, NY: Univ. at Buffalo.
- Delen, D., R. Sharda, and M. Bessonov. 2006. "Identifying significant predictors of injury severity in traffic accidents using a series of artificial neural networks." *Accid. Anal. Prev.* 38 (3): 434–444. <https://doi.org/10.1016/j.aap.2005.06.024>.
- Draper, N. R., and H. Smith. 1998. "Selecting the 'best' regression equation." In *Applied regression analysis*, 327–368. New York: Wiley.
- Eldin, N., and A. Senouci. 2011. "Condition rating of rigid pavements by neural networks." *Can. J. Civ. Eng.* 22 (5): 861–870. <https://doi.org/10.1139/195-104>.
- Ferreira, A., and L. P. Santos. 1999. "Pavement performance modelling: State of the art." In *Computer techniques for civil and structural engineering*. Edinburgh, UK: Edinburgh Civil-Comp Press.
- Fettahoglu, M. 2018. "Macroscopic state-level analysis of pavement roughness using time-space statistical modeling methods." M.S. thesis, Dept. of Civil, Structural, and Environmental Engineering, State Univ. of New York at Buffalo.
- FHWA (Federal Highway Administration). 2018. "Long-Term Pavement Performance." Accessed January 26, 2019. <https://highways.dot.gov/long-term-infrastructure-performance/ltp/p/long-term-pavement-performance>.
- Fountas, G., S. S. Pantangi, K. F. Hulme, and P. C. Anastasopoulos. 2019. "The effects of driver fatigue, gender, and distracted driving on perceived and observed aggressive driving behavior: A correlated grouped random parameters bivariate probit approach." *Anal. Methods Accid. Res.* 22 (Jun): 100091. <https://doi.org/10.1016/j.amar.2019.100091>.
- García de Soto, B., A. Bumbacher, M. Deublein, and B. T. Adey. 2018. "Predicting road traffic accidents using artificial neural network models." *Infrastruct. Asset Manage.* 5 (4): 132–144. <https://doi.org/10.1680/j/inam.17.00028>.
- Ghafoori, N., M. Najimi, J. Sobhani, and M. Aqel. 2013. "Predicting rapid chloride permeability of self-consolidating concrete: A comparative study on statistical and neural network models." *Constr. Build. Mater.* 44 (Jul): 381–390. <https://doi.org/10.1016/j.conbuildmat.2013.03.039>.
- Greene, W. H. 2011. *LIMDEP version 9 student reference guide*. New York: Econometric Software.
- Hegazy, T., P. Fazio, and O. Moselhi. 1994. "Developing practical neural network applications using back-propagation." *Comput.-Aided Civ. Infrastruct. Eng.* 9 (2): 145–159. <https://doi.org/10.1111/j.1467-8667.1994.tb00369.x>.
- Hong, F., and J. A. Prozzi. 2010. "Roughness model accounting for heterogeneity based on in-service pavement performance data." *J. Transp. Eng.* 136 (3): 205–213. [https://doi.org/10.1061/\(ASCE\)0733-947X\(2010\)136:3\(205\)](https://doi.org/10.1061/(ASCE)0733-947X(2010)136:3(205)).
- Hong, H. P., and S. Somo. 2001. "Probabilistic assessment of pavement conditions." *Can. J. Civ. Eng.* 28 (5): 813–822. <https://doi.org/10.1139/101-035>.
- InDoT (Indiana Department of Transportation). 2011. *Average daily traffic and commercial vehicles: Interactive map*. Indianapolis: InDoT.
- Khan, M. U., M. Mesbah, F. Luis, and J. W. David. 2014. "Developing a new road deterioration model incorporating flooding." *Transport* 167 (5): 322–333. <https://doi.org/10.1680/tran.13.00095>.
- Khattak, M. J., M. A. Nur, M. R. U. K. Bhuyan, and K. Gaspard. 2014. "International roughness index models for HMA overlay treatment of flexible and composite pavements." *Int. J. Pavement Eng.* 15 (4): 334–344. <https://doi.org/10.1080/10298436.2013.842237>.
- Kim, S. H., and N. Kim. 2006. "Development of performance prediction models in flexible pavement using regression analysis method." *KSCSE J. Civ. Eng.* 10 (2): 91–96. <https://doi.org/10.1007/BF02823926>.
- Labi, S., K. S. Hwee, G. Lamptey, and C. Nunoo. 2006. "Long-term benefits of microsurfacing applications in Indiana—Methodology and case study." In *Proc., 85th Annual Meeting of the Transportation Research Board*. Washington, DC: Transportation Research Board.
- La Torre, F., L. Domenichini, and M. I. Darter. 1998. "Roughness prediction model based on the artificial neural network approach." In *Proc., 4th Int. Conf. on Managing Pavements*. Washington, DC: Transportation Research Board.
- Lethanh, N., and B. T. Adey. 2013. "Use of exponential hidden Markov models for modelling pavement deterioration." *Int. J. Pavement Eng.* 14 (7): 645–654. <https://doi.org/10.1080/10298436.2012.715647>.
- Lou, Z., M. Gunaratne, J. J. Lu, and B. Dietrich. 2001. "Application of neural network model to forecast short-term pavement crack condition: Florida case study." *J. Infrastruct. Syst.* 7 (4): 166–171. [https://doi.org/10.1061/\(ASCE\)1076-0342\(2001\)7:4\(166\)](https://doi.org/10.1061/(ASCE)1076-0342(2001)7:4(166)).
- Luo, Z. 2013. "Pavement performance modelling with an auto-regression approach." *Int. J. Pavement Eng.* 14 (1): 85–94. <https://doi.org/10.1080/10298436.2011.617442>.
- Madanat, S., R. Mishalani, and W. H. Wan Ibrahim. 1995. "Estimation of infrastructure transition probabilities from condition rating data." *J. Infrastruct. Syst.* 1 (2): 120–125. [https://doi.org/10.1061/\(ASCE\)1076-0342\(1995\)1:2\(120\)](https://doi.org/10.1061/(ASCE)1076-0342(1995)1:2(120)).
- Mazari, M., and D. D. Rodriguez. 2016. "Prediction of pavement roughness using a hybrid gene expression programming-neural network technique." *J. Traffic Transp. Eng.* 3 (5): 448–455. <https://doi.org/10.1016/j.jtte.2016.09.007>.
- Moreira, A. V., J. Tinoco, J. R. Oliveira, and A. Santos. 2018. "An application of Markov chains to predict the evolution of performance indicators based on pavement historical data." *Int. J. Pavement Eng.* 19 (10): 937–948. <https://doi.org/10.1080/10298436.2016.1224412>.
- Nahidi, S., G. Fountas, S. P. Sarvani, M. T. Sarwar, and P. C. Anastasopoulos. 2017. "Project discrepancies in roadway construction and preservation: A statistical analysis of public–private partnership contract types in the US." *Front. Built Environ.* 3 (15): 1–20. <https://doi.org/10.3389/fbuil.2017.00015>.
- Plati, C., P. Georgiou, and V. Papavasiliou. 2016. "Simulating pavement structural condition using artificial neural networks." *Struct. Infrastruct. Eng.* 12 (9): 1127–1136. <https://doi.org/10.1080/15732479.2015.1086384>.
- Prozzi, J. A., and S. M. Madanat. 2003. "Incremental nonlinear model for predicting pavement serviceability." *J. Transp. Eng.* 129 (6): 635–641. [https://doi.org/10.1061/\(ASCE\)0733-947X\(2003\)129:6\(635\)](https://doi.org/10.1061/(ASCE)0733-947X(2003)129:6(635)).
- Rajagopal, A. 2006. *Developing pavement performance prediction models and decision trees for the city of Cincinnati*. Technical Rep. FHWA/OH-2006/14. Columbus, OH: Ohio Dept. of Transportation.
- Rose, S., B. S. Mathew, K. P. Isaac, and A. S. Abhaya. 2016. "Risk based probabilistic pavement deterioration prediction models for low volume roads." *Int. J. Pavement Eng.* 19 (1): 1–10. <https://doi.org/10.1080/10298436.2016.1162308>.
- Royston, J. P. 1982. "Algorithm AS 181: The W test for normality." *J. R. Stat. Soc. Ser. C: Appl. Stat.* 31 (2): 176–180. <https://doi.org/10.2307/2347986>.
- Saeed, T. U., T. Hall, H. Baroud, and M. J. Volovski. 2019. "Analyzing road crash frequencies with uncorrelated and correlated random-parameters count models: An empirical assessment of multilane highways." *Anal. Methods Accid. Res.* 23 (Sep): 100101. <https://doi.org/10.1016/j.amar.2019.100101>.
- Saeed, T. U., Y. Qiao, S. Chen, S. Alqadhi, Z. Zhang, S. Labi, and K. C. Sinha. 2017. *Effects of bridge surface and pavement maintenance activities on asset rating*. Joint Transportation Research Program, Publication No. FHWA/IN/JTRP-2017/19. West Lafayette, IN: Purdue Univ.
- Sandra, A. K., and A. K. Sarkar. 2013. "Development of a model for estimating international roughness index from pavement distresses." *Int. J. Pavement Eng.* 14 (8): 715–724. <https://doi.org/10.1080/10298436.2012.703322>.
- Sarwar, M. T., and P. C. Anastasopoulos. 2016. "Three-stage least squares analysis of postrehabilitation pavement performance." *Transp. Res. Rec.* 2589 (1): 97–109. <https://doi.org/10.3141/2589-11>.
- Sarwar, M. T., and P. C. Anastasopoulos. 2017. "The effect of long-term non-invasive pavement deterioration on accident injury-severity rates: A seemingly unrelated and multivariate equations approach." *Anal. Methods Accid. Res.* 13 (Mar): 1–15. <https://doi.org/10.1016/j.amar.2016.10.003>.
- Sarwar, M. T., A. Bhargava, S. S. Pantangi, P. C. Anastasopoulos, and S. B. Mohan. 2019. *Public-private partnerships for roadway construction and preservation, and traffic safety: An exploratory empirical analysis of crash frequencies*. No. 19-01336. Washington, DC: Transportation Research Board.

- Silva, F., T. Van Dam, W. Bulleit, and R. Ylitalo. 2000. "Proposed pavement performance models for local government agencies in Michigan." *Transp. Res. Rec.* 1699 (1): 81–86. <https://doi.org/10.3141/1699-11>.
- Sobhani, J., M. Najimi, A. R. Pourkhorshidi, and T. Parhizkar. 2010. "Prediction of the compressive strength of no-slump concrete: A comparative study of regression, neural network and ANFIS models." *Constr. Build. Mater.* 24 (5): 709–718. <https://doi.org/10.1016/j.conbuildmat.2009.10.037>.
- Sollazzo, G., T. F. Fwa, and G. Bosurgi. 2017. "An ANN model to correlate roughness and structural performance in asphalt pavements." *Constr. Build. Mater.* 134 (Mar): 684–693. <https://doi.org/10.1016/j.conbuildmat.2016.12.186>.
- Terzi, S. 2007. "Modeling the pavement serviceability ratio of flexible highway pavements by artificial neural networks." *Constr. Build. Mater.* 21 (3): 590–593. <https://doi.org/10.1016/j.conbuildmat.2005.11.001>.
- Waseem, M., A. Ahmed, and T. U. Saeed. 2019. "Factors affecting motorcyclists' injury severities: An empirical assessment using random parameters logit model with heterogeneity in means and variances." *Accid. Anal. Prev.* 123 (Feb): 12–19. <https://doi.org/10.1016/j.aap.2018.10.022>.
- Washington, S. P., M. G. Karlaftis, and F. Mannering. 2011. *Statistical and econometric methods for transportation data analysis*. 2nd ed. Boca Raton, FL: Chapman and Hall/CRC Press.
- Xu, G., L. Bai, Z. Sun, T. Nowaczyk, C. Shive, and J. Wilcoxson. 2015. "Pavement deterioration prediction model and project selection for Kentucky highways." In *Proc., 9th Int. Conf. on Managing Pavement Assets (ICMPA9)*. Alexandria, VA: Virginia Tech Transportation Institute.
- Yamany, M. S., and E. Elwakil. 2019. "Modeling of cast-in-place concrete tunnel liners condition." *Struct. Infrastruct. Eng.* <https://doi.org/10.1080/15732479.2019.1687529>.
- Yamany, M. S., T. U. Saeed, and M. Volovski. 2019. *Performance prediction of interstate flexible pavement across the Midwestern United States: Random-parameter regression vs artificial neural network*. No. 19-02723. Washington, DC: Transportation Research Board.
- Yang, J., J. Lu, M. Gunaratne, and Q. Xiang. 2003. *Forecasting overall pavement condition with neural networks: Application on Florida highway network*. Washington, DC: Transportation Research Board.
- Zhang, Q. J., and K. C. Gupta. 2000. *Neural networks for RF and microwave design*. Norwood, MA: Artech House.
- Ziari, H., J. Sobhani, J. Ayoubinejad, and T. Hartmann. 2016. "Prediction of IRI in short and long terms for flexible pavements: ANN and GMDH methods." *Int. J. Pavement Eng.* 17 (9): 776–788. <https://doi.org/10.1080/10298436.2015.1019498>.